

Algorithmic Choice in Bluesky

Responsible ranking + security in an open feed ecosystem

Measured: discovery → panel → 600,651 ranked impressions (Feb 1–2, 2026)

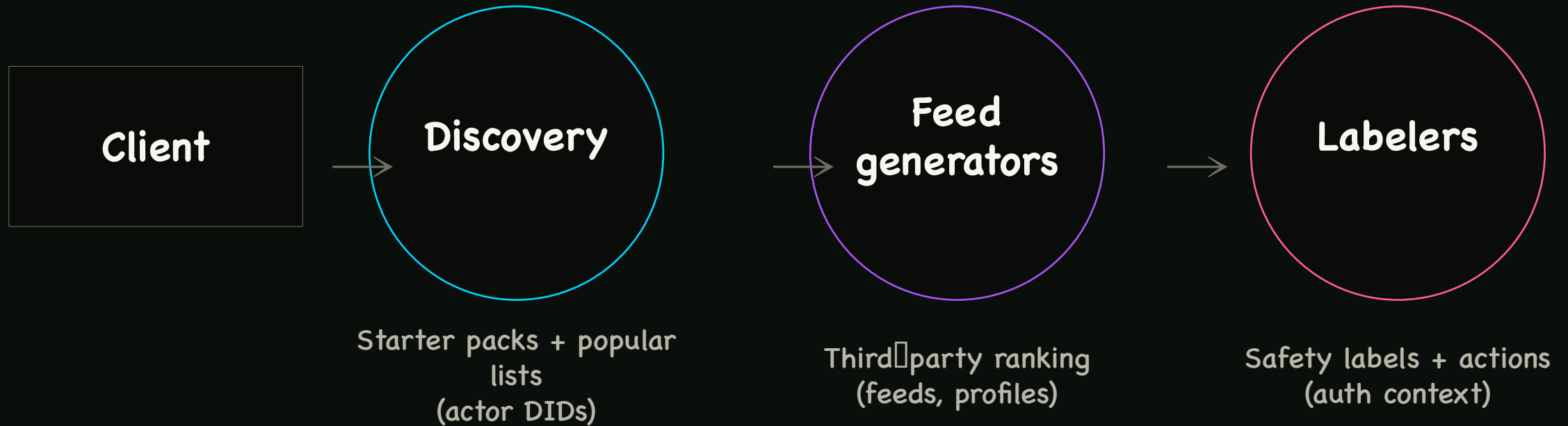
Algorithmic choice ≠ decentralized power

- Feeds + labelers turn ranking/moderation into a marketplace.
- Discovery + hosting create chokepoints (and targets).
- New attack surface: malicious feeds, growth gaming, label gaps, privacy.

Thesis: pluralism shifts power to discovery + infrastructure.

Bluesky's open feed ecosystem

Ranking + moderation become third-party services (a marketplace).



Chokepoints emerge where users find feeds + who hosts them. [ICWSM'24 10.1609/icwsm.v18i1.31293]

Discovery is measurable (not anecdotes)

Starter packs + popular feeds are explicit lists that seed attention.

Starter packs (onboarding bundles)

Starter pack (metadata)

Name	Creator DID	Pack URI
Lauren Raymor's Starter Pack	did:plc:dwzj...b7cawl	.../app.bsky.graph.starterpack/3mdto57g

Example row (truncated)

Starter pack → feed inclusion

Pack	Feed
.../app.bsky.graph.starterpack/3mdto57gqjd2g	.../app.bsky.feed.generator/mich-dot-gov

Example row (truncated)

Popular feeds (in-app list)

Popular feeds (example row)

Rank	Feed	URI
top	Discover	.../app.bsky.feed.generator/whats-hot

Example row (truncated)

These surfaces seed attention → panel sampling. [ICWSM'25 10.1609/icwsm.v19i1.35810]

What we measured (H1-H6)

Discovery → provider leverage → exposure → overlap → rich-get-richer → safety labels

H1 Discovery concentration

Do a few feeds dominate starter-pack inclusion?

H2 Provider leverage

Which providers get outsized discovery?

H3 Exposure concentration

How concentrated is exposure?

H4 Feed overlap

Do feeds show the same posts?

H5 Exposure vs author size

Do larger authors get outsized exposure?

H6 Risk exposure variability (labels)

Problem: responsible ranking meets open markets

- Users choose algorithms — but discovery chooses defaults.
- Concentration shifts attacks to discovery + providers.
- Responsible ranking needs: security, fairness, safety, accountability.

Threat model: marketplace adversaries

- Malicious feed: steers exposure, harassment, or explicit content.
- Growth-gaming: coordinated engagement to pump accounts.
- Labeler gaps: evasion, inconsistent semantics, and bias.

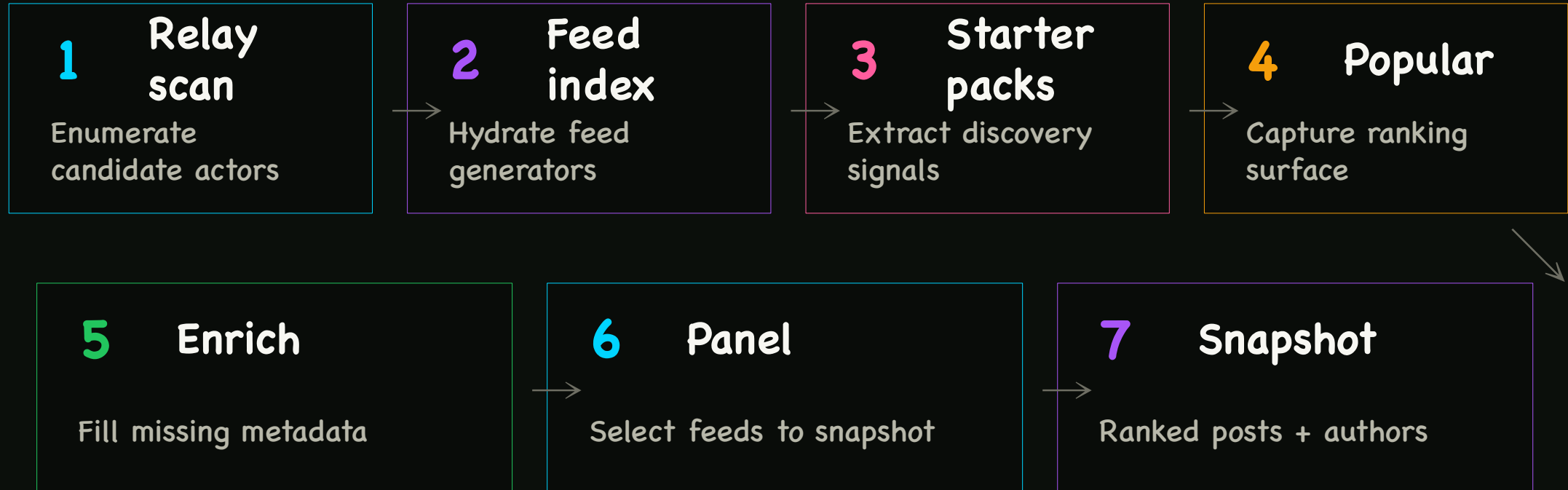
Cites: NDSS'21 10.14722/ndss.2021.24525; USENIX'23 10.5555/3620237.3620333

Our contributions (S&P tone)

- Threat model for ranking+moderation marketplaces.
- Reproducible measurement: discovery → panel → impressions.
- Evidence + mitigations for concentration, leverage, overlap, and risk.

Methods credibility (read-only snapshot)

Discovery → fixed panel → ranked impressions → joinable tables



All calls are read-only XRPC (run shown: unauth / public AppView).

The run behind this story

Run settings (minimal)

viewer: unauth (public AppView)
window: 2026-02-01T09:04:54Z → 2026-02-02T08:23:24Z
run_id: 88b597ae...bd1967

Anchors (2 numbers)

Panel size: 6,000 feeds
Captured: 600,651 ranked impressions

Every stage leaves a trace (exports)

- Feed generator index
- Starter packs + inclusions
- Popular feed list
- Feed panel
- Ranked impressions
- Posts (metadata) + authors

Feed generator index

Feed	Provider	URI
Discover	discover.bsky.app	.../app.bsky.feed.generator/whats-hot

Example row (truncated)

(example: one row from the index)

Joinable dataset (keys line up)

Panel → Impression → Post metadata → Author profile (no post text).

Panel entry

Feed panel (chosen feed)

Feed	Group	Chosen by	URI
Discover	Popular	Top of popular list	.../app.bsky.feed.generator/wh

Example row (truncated)

Ranked impression

Ranked impression (example)

Viewer	Rank	Post	Author DID
unauth (public)	top	.../app.bsky.feed.post/3mdtd7svxgr22su...hu6tln	did:plc:22su...hu6tln

Example row (truncated)

Post metadata (no text)

Post metadata (no text)

Post	Embed	Images	External
.../app.bsky.feed.post/3mdtd7svxgr22su...hu6tln	images	yes	(none)

Langs
["en"]

Example row (truncated)

Author profile

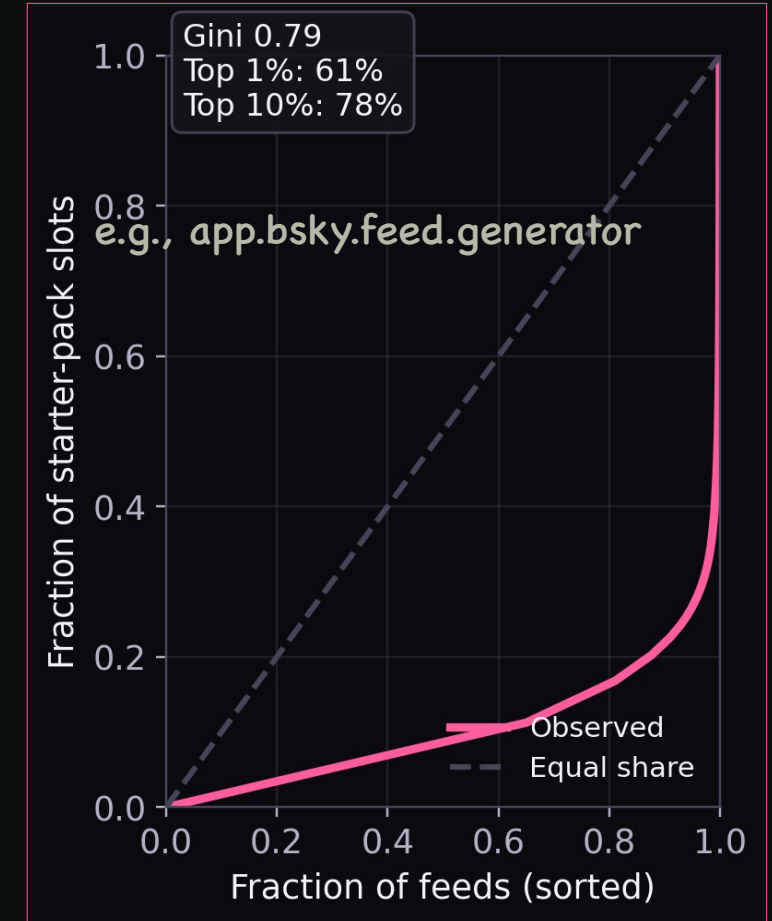
Author profile (no counts)

Handle	Name	DID
landguyminor.bsky.social	Landguy	did:plc:22su...hu6tln

Example row (truncated)

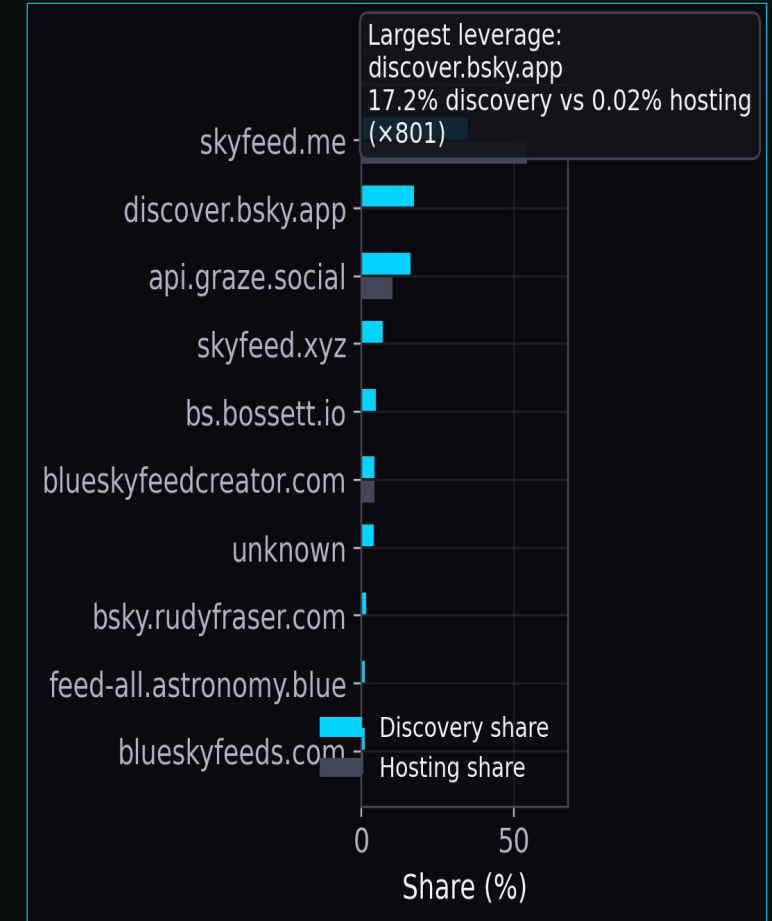
H1 — Discovery slots are concentrated

- Starter-pack slots seed default attention.
- Top 1%: 61% of slots (Gini 0.79).
- Discovery becomes a chokepoint.



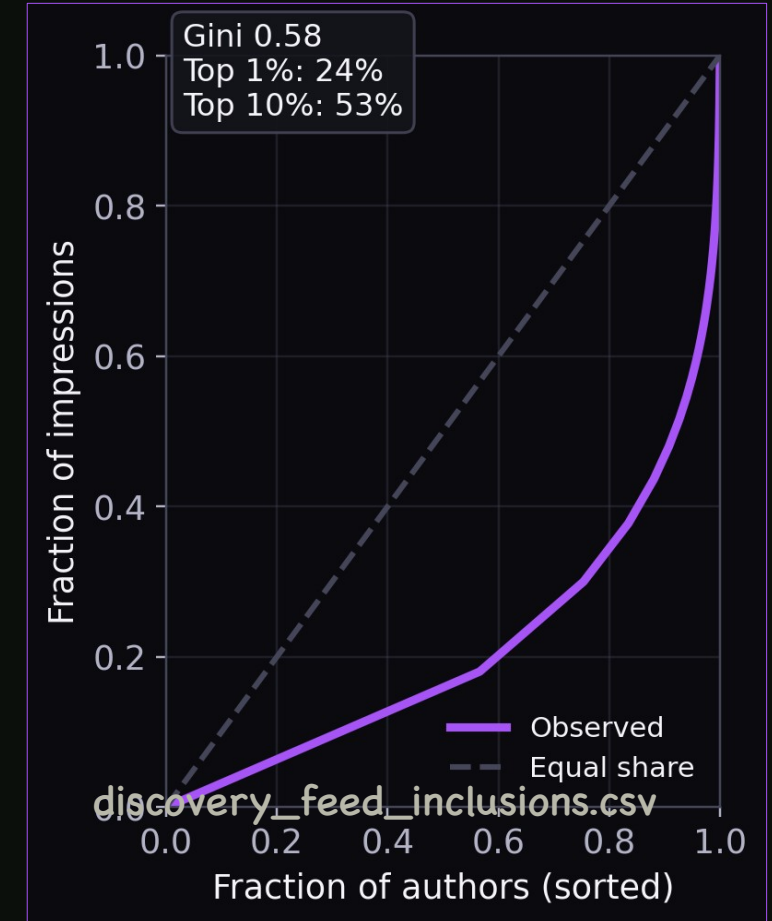
H2 — Discovery creates provider leverage

- Hosting providers form a feed 'supply chain'.
- 0.02% hosting → 17% discovery (×801).
- Gatekeepers become targets.



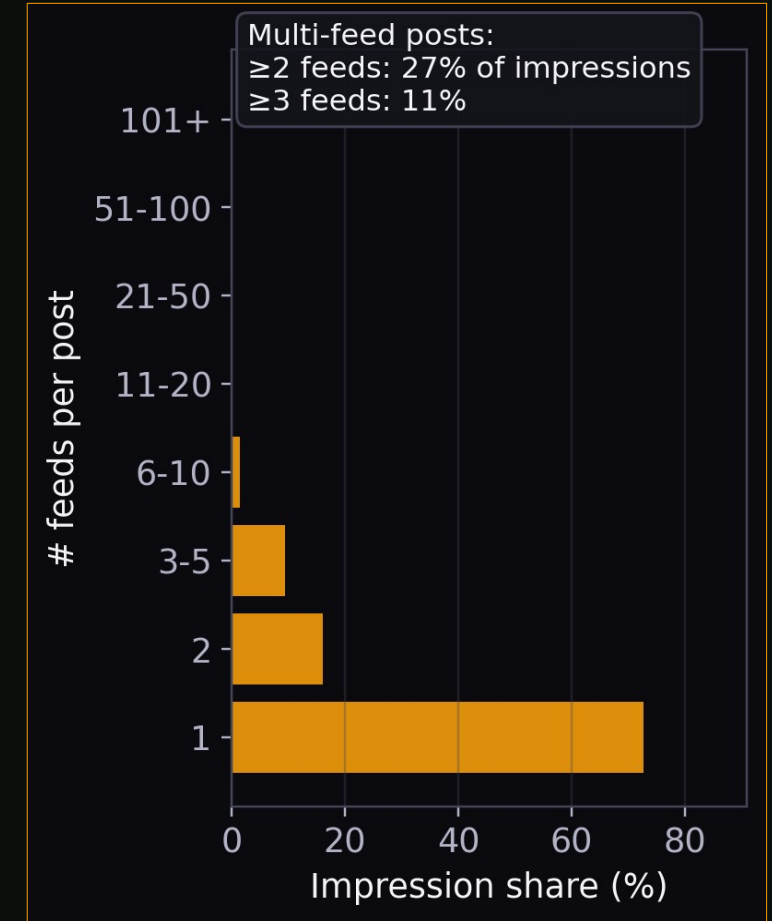
H3 — Exposure is concentrated among authors

- Ranked feeds amplify a small creator set.
- Top 10% authors: 53% of impressions (Gini 0.58).
- Cheaper manipulation + targeting.



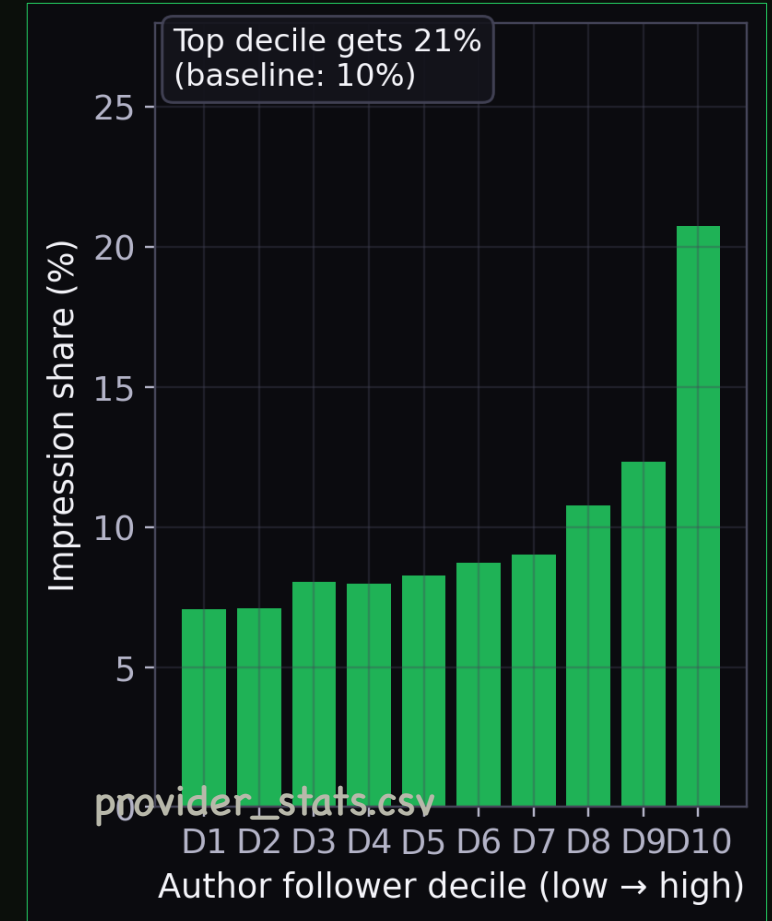
H4 — Many feeds show the same posts

- Pluralism $\neq$ diversity by default.
- 27% of impressions are posts in ≥ 2 feeds.
- Redundancy amplifies winners.



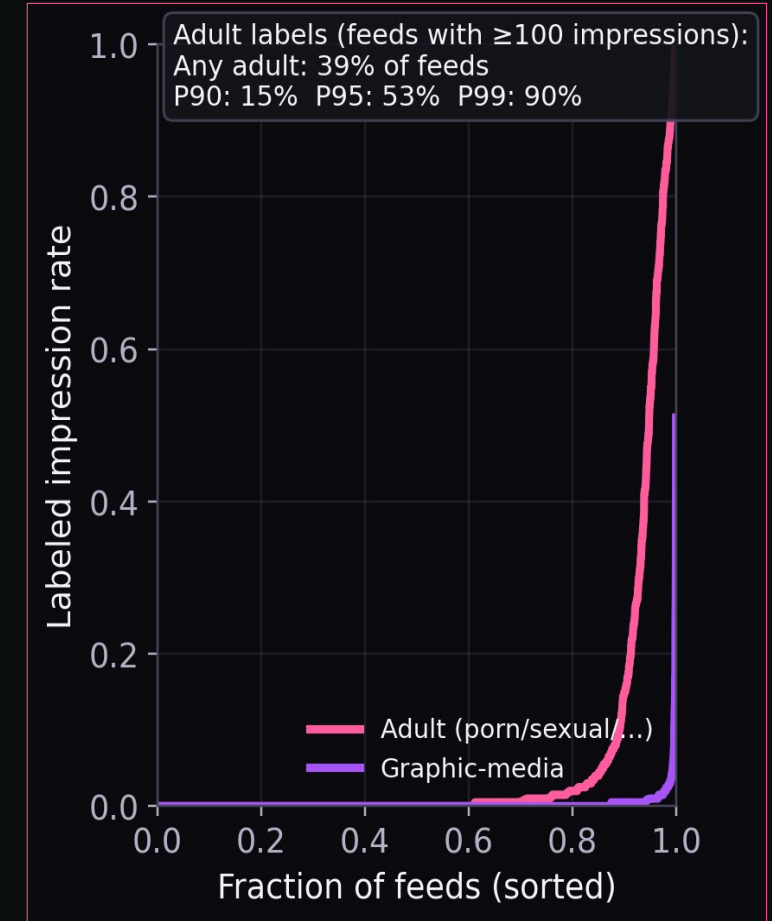
H5 — Bigger accounts get outsized exposure

- Exposure tilts toward large accounts.
- Top follower decile gets 21% (baseline 10%).
- Popularity bias reappears.



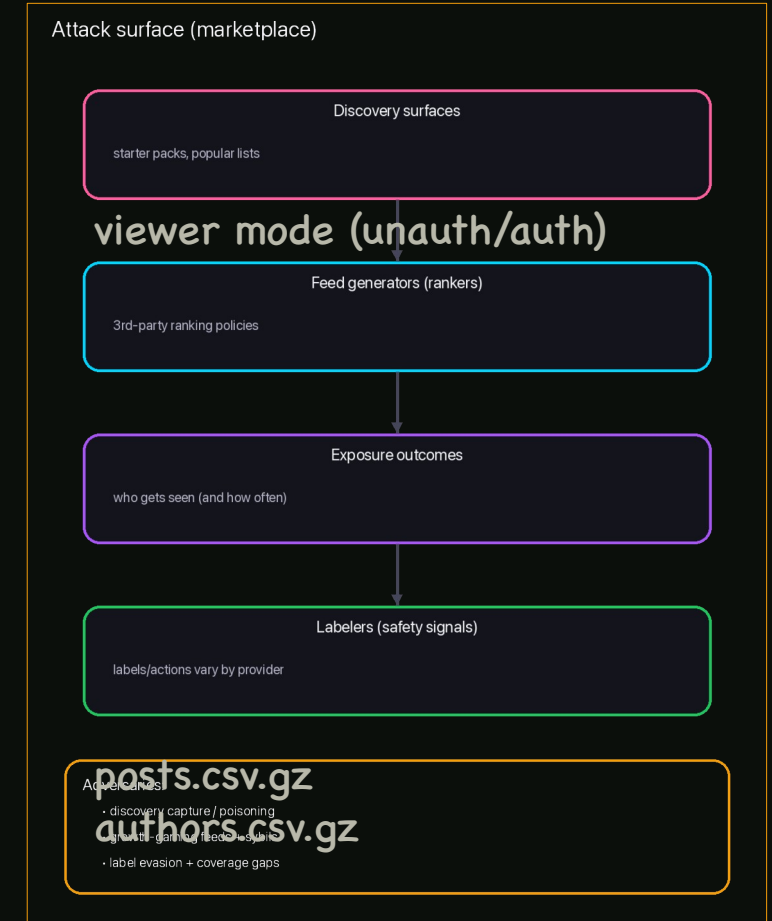
H6 — Safety label exposure varies by feed

- Feeds differ in labeled-risk they surface.
- Feeds (≥ 100 impressions): 39% any adult; P95 rate=53%.
- Risk boundaries depend on feed choice.



Security implications: a new attack surface

- Concentrated discovery reduces attack cost.
- Third-party rankers can be adversarial by design.
- Defenses must be marketplace-wide (audit + guardrails).



Privacy: feeds & labelers can learn about you

Thirdparty services can profile from request patterns + outputs.

GET

`https://.../xrpc/app.bsky.feed.getFeed?feed=at://...&cursor=...`

Host

What a service can observe:

- IP + timing + feed URI
- (if auth) viewer DID + behavior
- repeated queries reveal preferences
- AppView (public): feeds, posts, profiles
- Relay: network index for enumeration

Method

Why it matters:

- profiling / tracking
- output leakage (membership inference)
- cross-service linkage risk

Example: `app.bsky.feed.getFeed` = “return a ranked feed view”.

(We keep slides non-code: method names are the “API surface”.)

Implication: decentralized moderation is fragile

- Label semantics are subjective → disagreement across labelers.
- Bias + coverage gaps are documented in toxicity labeling.
- Feed choice can bypass or amplify safety boundaries.

Cites: NAACL'22 10.18653/v1/2022.naacl-main.431; Findings ACL'22 10.18653/v1/2022.findings-acl.176

Mitigation: discovery transparency + diversification

- Publish logs for starter packs + popular lists (auditability).
- Diversify onboarding defaults; cap single-feed dominance.
- Detect manipulation (suspicious growth + overlap spikes).

Mitigation: auditable feed marketplace

Logs + reproducible snapshots enable accountability

Add audit hooks around discovery, ranking, and labelers.

Expose stable identities + policies so clients can warn and researchers can audit.

Transparency logs

Audit APIs

Reproducible snapshots

Why this matters

Choice only helps if users can compare risk, diversity, and trust. auditable, and easy to analyze.

What to disclose (minimal)

- provider identity + hosting footprint
- discovery surfaces (who promoted what)
- ranking output summaries (exposure / overlap)
- labeler policies + coverage
- incident response + takedown hooks

Mitigation: “feed nutrition labels” (user-facing)

- Show provider, overlap, and exposure concentration at a glance.
- Show label exposure rates + which labelers are applied.
- Safer interfaces: warnings, filters, and defaults.

Mitigation: labeler interoperability + secure sharing

- Standardize vocabularies + transparency for labeler policies.
- Share signals privately (limit linkage) + resist poisoning.
- Encourage multi-labeler consensus (reduce single-point failures).

Cites: IEEE S&P'22 10.1109/SP46214.2022.9833647

Takeaways

- Algorithmic choice shifts power to discovery + infrastructure.
- Open markets create new adversarial surfaces + inequalities.
- We provide evidence + threat model + mitigations (data/code).

Questions?

References (venue-filtered, 2021–2025)

CCS / IEEE S&P / NDSS / USENIX Sec / WWW / CSCW / ICWSM / ACL / KDD

Decentralization + discovery

ICWSM'25 Starter Packs
<https://doi.org/10.1609/icwsm.v19i1.35810>

ICWSM'24 Decentralised Moderation
<https://doi.org/10.1609/icwsm.v18i1.31293>

ICWSM'24 Group Sanctions
<https://doi.org/10.1609/icwsm.v18i1.31316>

WWW'23 Will Admins Cope?
<https://doi.org/10.1145/3543507.3583487>

CSCW'24 Trouble in Paradise
<https://doi.org/10.1145/3687059>

Ranking + security + labels

NDSS'21 Poisoning Recommenders
<https://doi.org/10.14722/ndss.2021.24525>

USENIX'23 PORE
<https://doi.org/10.5555/3620237.3620333>

CCS'21 Membership Inference
<https://doi.org/10.1145/3460120.3484770>

IEEE S&P'22 FL Poisoning Eval
<https://doi.org/10.1109/SP46214.2022.9833647>

KDD'21 Popularity Bias (Dynamic)
<https://doi.org/10.1145/3447548.3467376>

NAACL'22 Annotators w/ Attitudes
<https://doi.org/10.18653/v1/2022.naacl-main.431>

Findings ACL'22 Fairness Varies
<https://doi.org/10.18653/v1/2022.findings-acl.176>

Findings EMNLP'23 Hate Bias (Causal)
<https://doi.org/10.18653/v1/2023.findings-emnlp.440>